UART, I²C, AND SPI

SERIAL COMMUNICATION PROTOCOL

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TOPIC OUTLINE

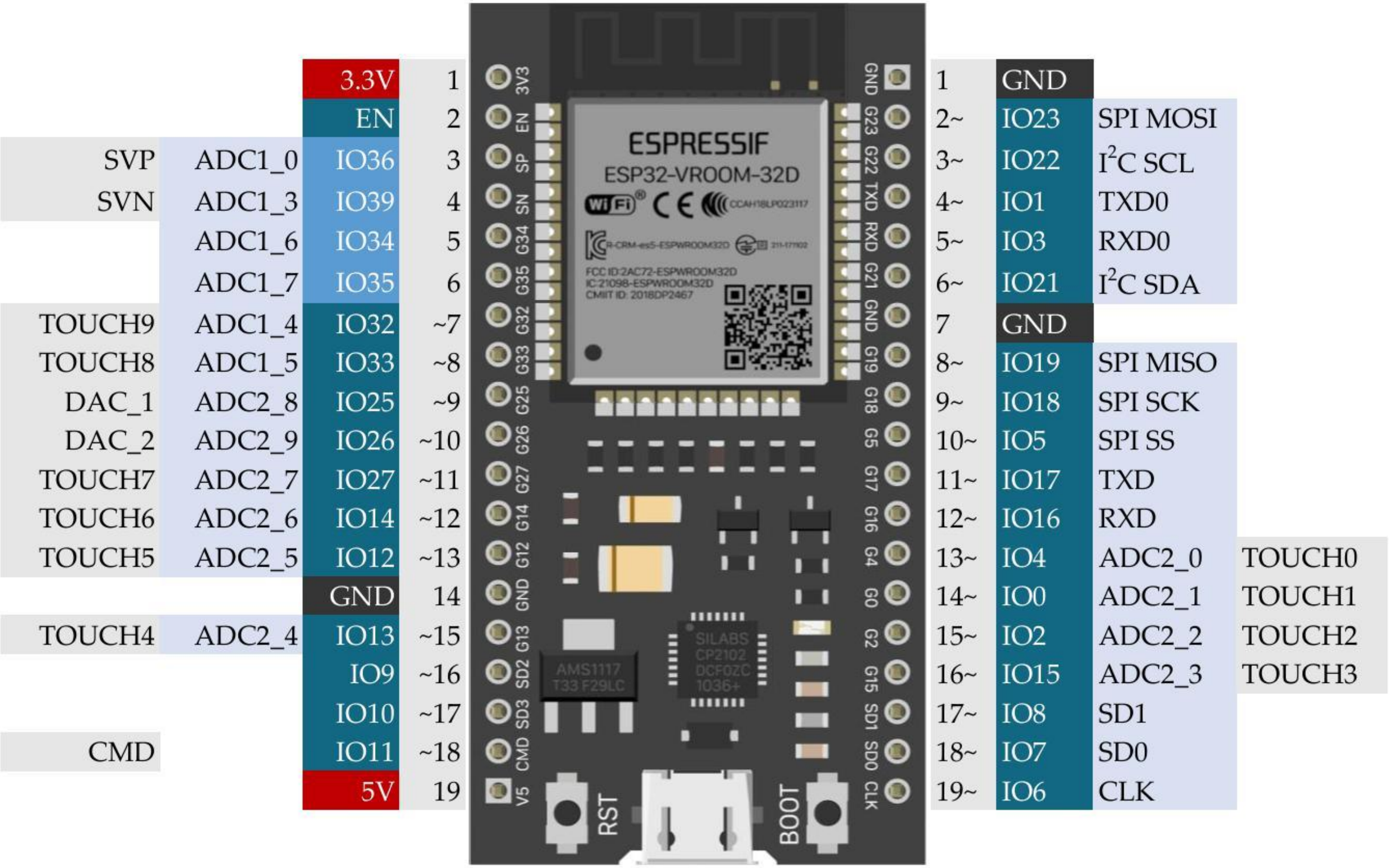
UART

I^2C

SPI

UART

DEVKITC GPIOs MAP



GPIO voltage: 3.3V (NOT 5V tolerant)

GPIO current: 12-20 mA (40 mA max)

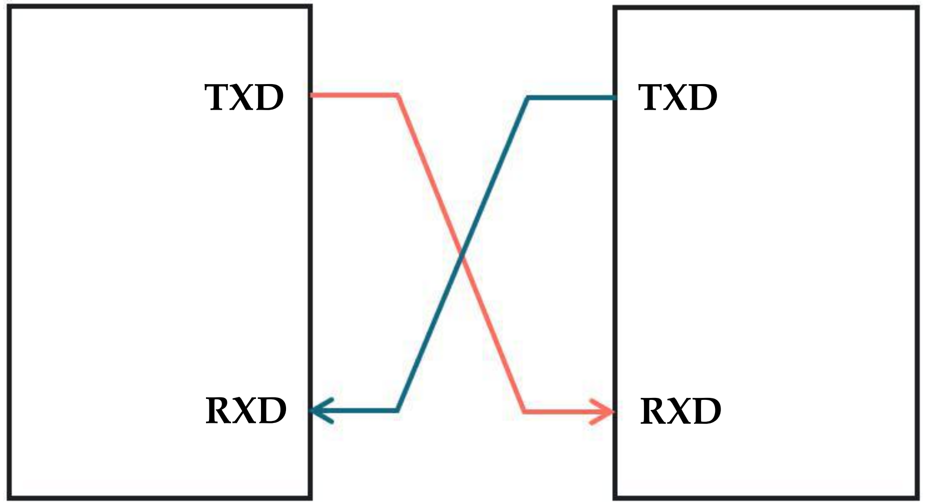
[ESP32-DevKitC-1 - - — Arduino ESP32 latest documentation]

UART

<u>UART</u> (**Universal Asynchronous Receiver-Transmitter**) is a serial communication method that sends and receives data one bit at a time between two devices. It does not require a clock signal because both devices agree on a common baud rate (data transmission speed).

Asynchronous

- TX and RX do not share a common clock

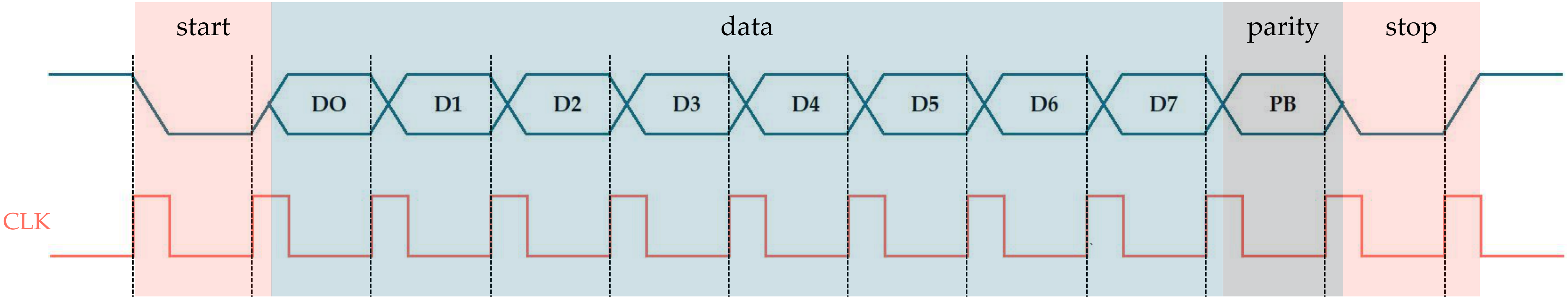


Baud rates

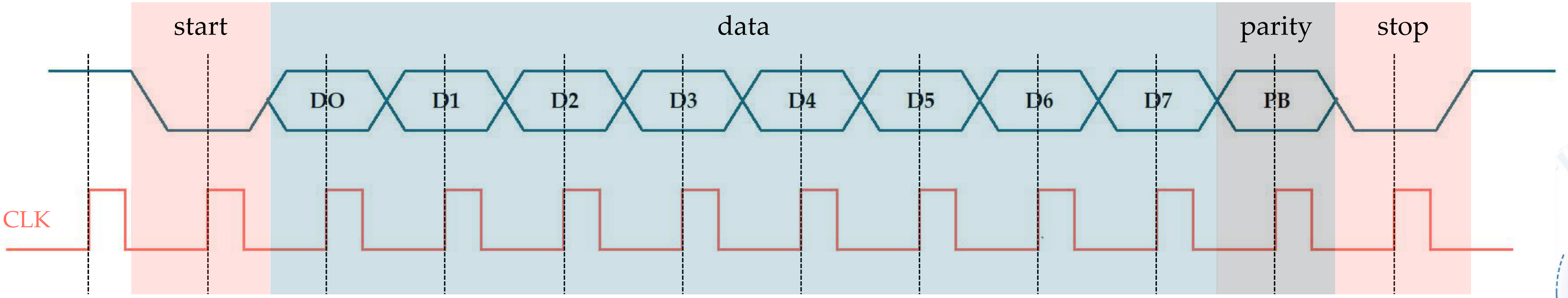
- 4800
- 9600
- 19200
- 57600
- 115200

UART FRAME FORMAT

Transmitter



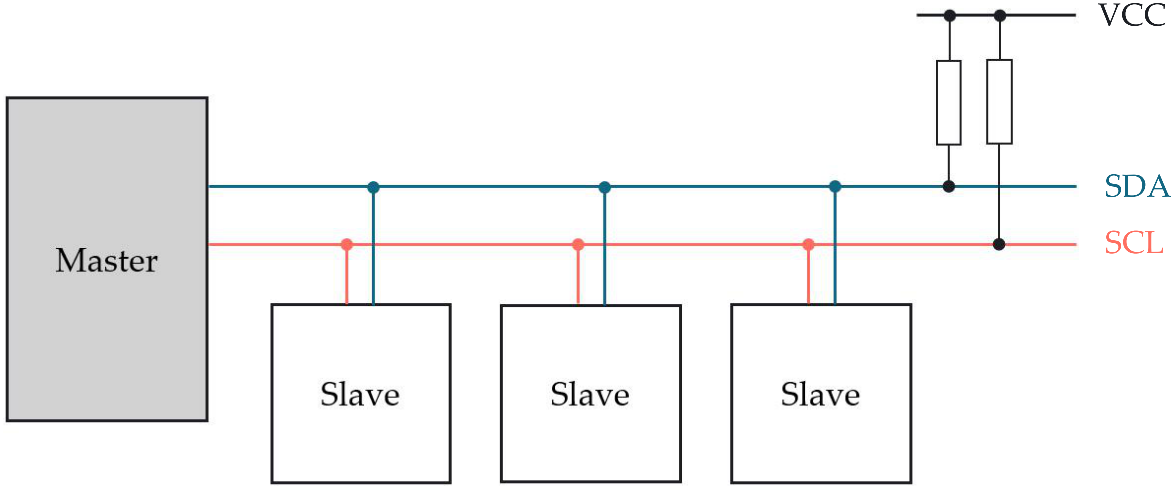
Receiver



I^2C

I^2C

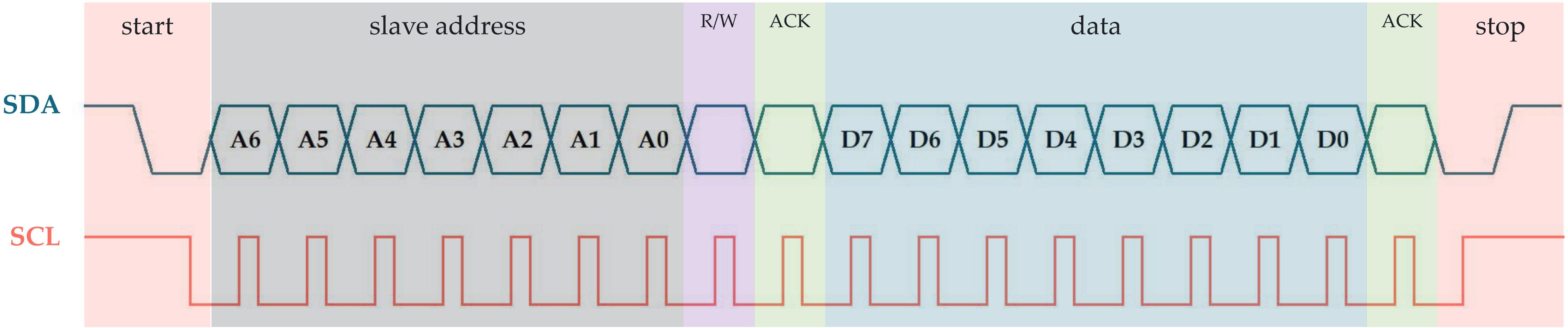
Basic I^2C Topology



I^2C (**Inter-Integrated Circuit**) is a serial communication protocol that allows multiple devices to communicate using only two wires: one for **data** (SDA) and one for the **clock** signal (SCL). It is commonly used to connect sensors, displays, and microcontrollers.

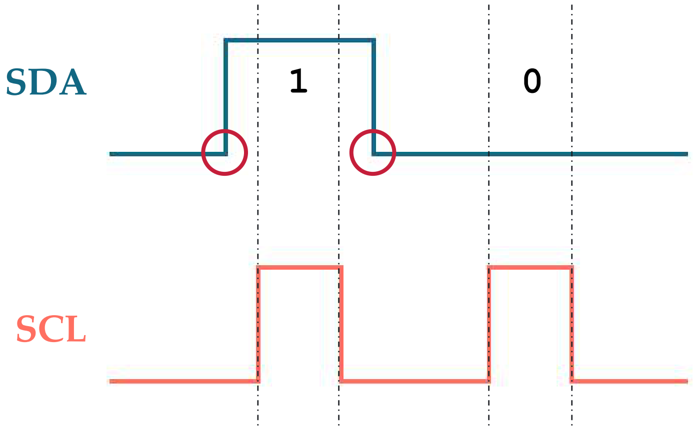
The master selects and controls a slave device using its unique address.

I²C FRAME FORMAT



Timing relationship between SDA and SCL

- During data transmission, SDA only transitions while SCL is low

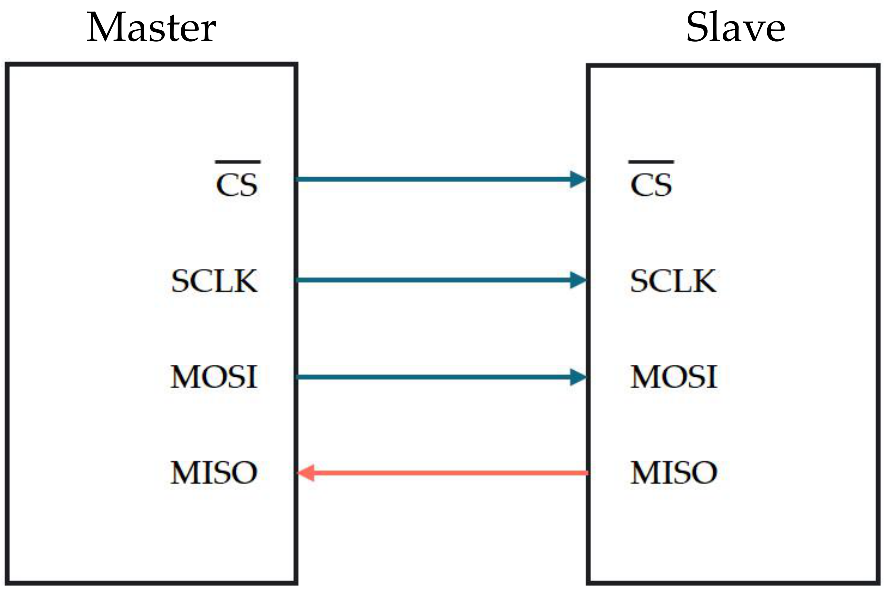


SPI

SPI

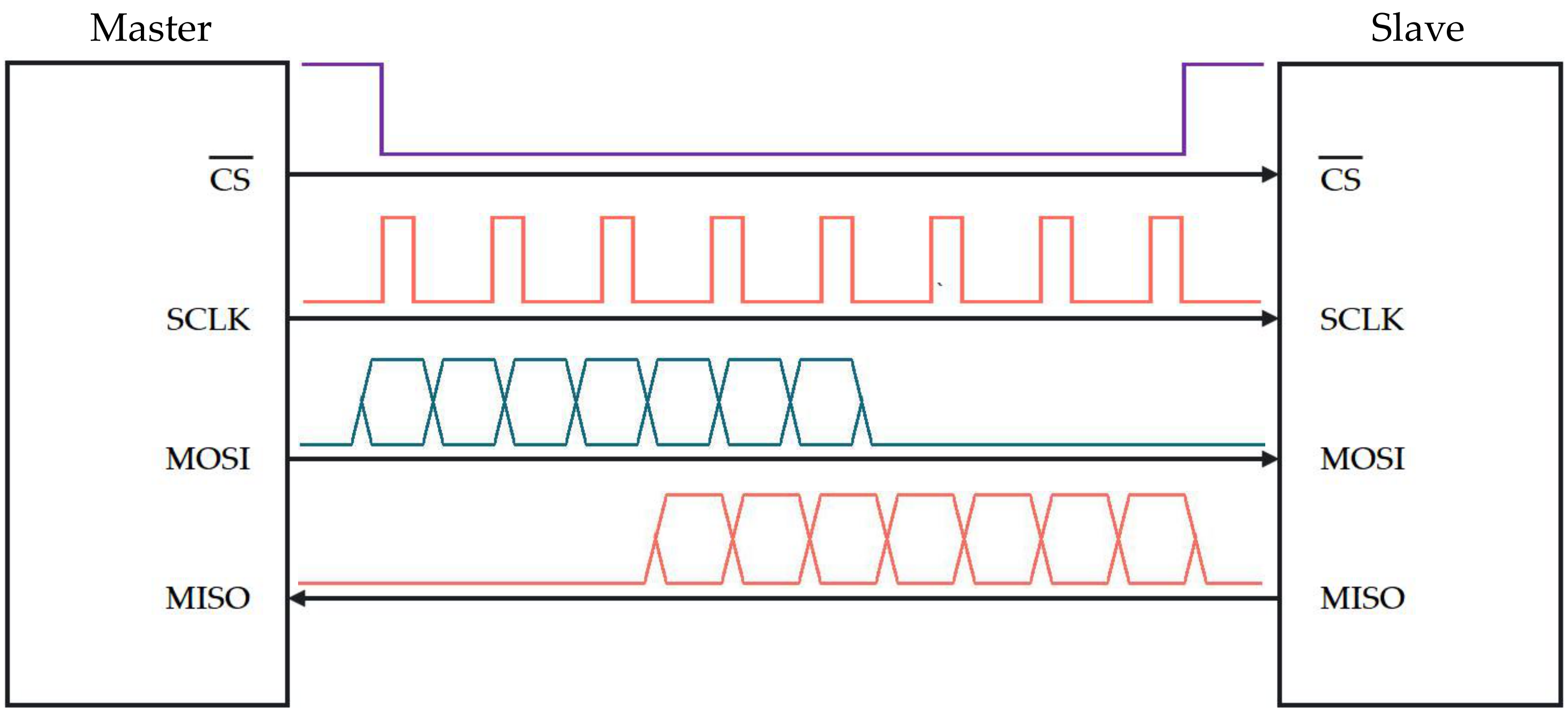
[SPI]() (**Serial Peripheral Interface**) is a high-speed serial communication protocol used for communication between a master device and one or more slave devices. It uses separate lines for data transmission, data reception, clock, and device selection.

SPI Nomenclature



- **CS** (Chip Select)
- **SCLK** (Synchronous Clock)
- **MOSI** (Master Out Slave In)
- **MISO** (Master In Slave Out)

OVERVIEW OF SPI PROTOCOL



CS - Chooses communication target

SCLK - Provides timing/synchronization

MOSI - Data transmitted by master

MISO - Data received by master

SUMMARY

Feature	UART	I2C	SPI
Communication Type	Asynchronous	Synchronous	Synchronous
Number of Wires	2 (TX, RX)	2 (SDA, SCL)	4 or more (MOSI, MISO, SCLK, SS)
Number of Devices	Usually 2 devices	Multiple devices	Multiple Devices
Communication Speed	Moderate	Moderate	High
Device Addressing	No addressing	Uses addresses	Uses slave-select lines
Complexity	Simple	Moderate	Moderate to High
Common Applications	Computer-to-microcontroller communication, GPS modules	Sensors, RTCs, EEPROMs	SD cards, displays, high-speed peripherals

LABORATORY